Deep Learning in Computer Vision· Lecture 8 of 9

Object Detection Basics

Boxes, IoU, anchors, non-max suppression (NMS), mAP and the road to YOLO

Where We Are, and Roadmap

- **Classification**: what is in the image.
- **Segmentation**: which pixels.
- **Detection**: **what and where, for many objects**: a box, a class and a score for each.
    - **Part A**: the task, box formats, and **IoU**, the tool for measuring overlap.
    - **Part B**: how detectors evolved: sliding window → R-CNN → Fast → Faster R-CNN, and **anchors**.
    - **Part C**: one-stage detectors and **non-maximum suppression (NMS)**.
    - **Part D**: evaluation with **AP / mAP** and datasets.

Glossary: Terms Used in This Lecture

Term	Meaning
Bounding box	Axis-aligned rectangle around an object, given by 4 numbers
Class label	Category of the object (person, car, …)
Confidence score	Detector's certainty (0-1) that the box contains an object of that class
Ground truth (GT)	Human-annotated correct boxes and labels
IoU	Intersection over Union: overlap measure between two boxes
Region proposal / RoI	A candidate box that may contain an object / region of interest
Anchor	Predefined reference box (fixed size and shape) that the network corrects
NMS	Non-maximum suppression: deletes duplicate detections
Regression	Predicting continuous numbers (here: box coordinates)
TP / FP / FN	True positive / false positive / false negative detection
AP / mAP	Average Precision of one class / mean Average Precision over classes

From Classification to Detection

Task	Question answered	Output
Classification	What is in the image?	1 label
Localization	What is it and where? (one object)	1 label + 1 box
Detection	**What and where, for all N objects?**	**N × (label, box, score)**; **N unknown**

- Detection is harder because the **number of outputs varies** with the image: 0, 3 or 30 objects.
- A box is a **tight rectangle** around the visible object; it contains some background.
- Applications: driving (cars, pedestrians), retail (products), industry (defects), medicine (cells, lesions), security, wildlife monitoring.

What a Detector Outputs



Class	Score	Box (x1, y1, x2, y2)
person	0.97	(120, 40, 280, 400)
person	0.88	(300, 60, 420, 410)
dog	0.74	(450, 250, 600, 420)

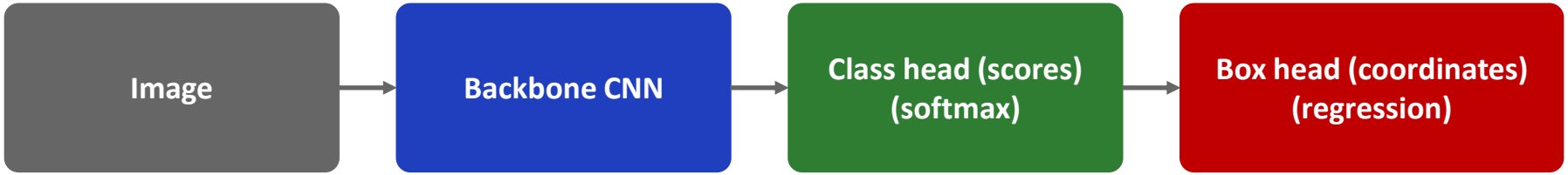
- **Score** = confidence (0-1); a **threshold** (e.g. 0.5) removes weak detections.
- Coordinates are in **pixels**; (x1, y1) is the **top-left** corner and (x2, y2) the **bottom-right** (origin is the image top-left, y grows downward).

Bounding Box Formats and Conversion

Format	Values	Used by
Corners	(x_1, y_1, x_2, y_2) or $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$	Pascal VOC, many libraries
Corner + size	(x, y, w, h)	COCO
Centre + size	$(\boldsymbol{c_x}, c_y, w, h)$	YOLO
Normalized	values divided by image width / height → [0, 1]	YOLO label files

- **Conversion**: $\boldsymbol{c_x} = (x_1 + x_2)/2$, $\boldsymbol{c_y} = (y_1 + y_2)/2$, $w = x_2 - x_1$, $h = y_2 - y_1$.
- **Example** (image 640×480, box (160, 120, 320, 360)):
  - $\boldsymbol{c_x}$ = 240, $\boldsymbol{c_y}$ = 240, w = 160, h = 240.
  - **Normalized**: $\boldsymbol{c_x}$ = 240/640 = **0.375**, $\boldsymbol{c_y}$ = 240/480 = **0.5**, w = 160/640 = **0.25**, h = 240/480 = **0.5**.
- **Always check which format and units a dataset or library uses; mixing them is a classic bug.**

Detection = Classification + Regression



- **Backbone**: a CNN (ResNet, VGG …) that extracts features.
- **Class head**: scores for each class, trained with cross-entropy (**classification**).
- **Box head**: predicts continuous numbers (x, y, w, h);
  - this is **regression**, trained with a distance loss such as **L1** $|x - \hat{x}|$ or **IoU-based** losses.
- **Total loss** = $L_{cls} + \lambda \cdot Lbox$
  - λ balances the two terms.
- Difficulty: the **number of boxes** is not fixed, so detectors first generate many **candidate** locations.

Why Detection is Harder

Many outputs

Variable number of objects, so the output size is not fixed.

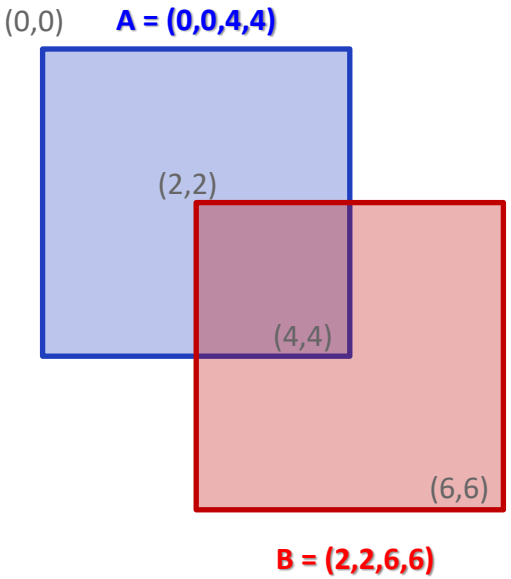
Scale, shape, occlusion

Objects range from tiny to image-sized, different aspect ratios, partly hidden.

Extreme imbalance

A detector looks at ~10^4-10^5 candidate locations but only a handful contain objects. Almost all are **background** (class imbalance).

Intersection over Union (IoU)



$$IoU = area(A \cap B) / area(A \cup B)$$

Symbol / term	What it means
$A \cap B$	intersection: area covered by both boxes
$A \cup B$	union: area covered by either box = area(A) + area(B) − area($A \cap B$)
IoU = 1	boxes identical
IoU = 0	boxes do not overlap
Example	intersection 2×2 = 4; union 16 + 16 − 4 = 28; **IoU = 4/28 ≈ 0.14**

IoU as a Decision Rule: TP, FP, FN

- A **match** needs the right **class** and **IoU ≥ a threshold** (usually **0.5**).
- Detections are processed in **descending score** order;
  - **True positive (TP)**: a detection that matches a ground-truth (GT) box not yet matched.
  - **False positive (FP)**: wrong class, IoU below the threshold, or a **duplicate** of an already matched object.
  - **False negative (FN)**: a ground-truth object with no matching detection.
  - Each GT box can be matched **once**.
- These counts feed:
  - precision = $\frac{TP}{(TP+FP)}$ and
  - recall = $\frac{TP}{(TP+FN)}$

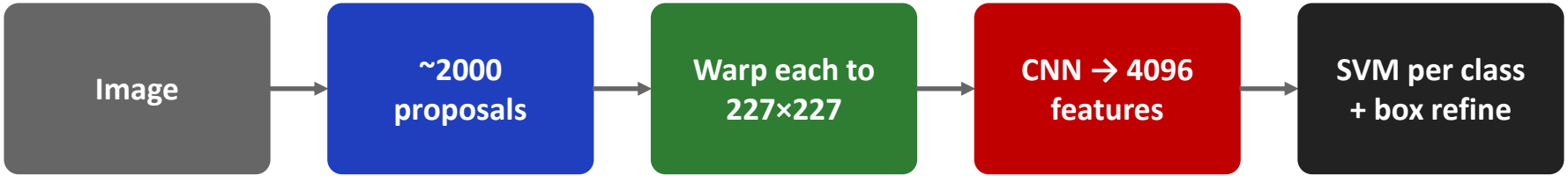
Naive Idea: Sliding Window

- **Slide a fixed window** over the image and **classify each crop with a CNN**.
  - Crop = one candidate image window
- **Example:** 640×480 image, 64×64 window, **stride 8** (step between windows):
  - $\left(\frac{(640-64)}{8}+1\right)\times\left(\frac{(480-64)}{8}+1\right)=73\times53=$ **3,869 windows**.
- Objects differ in size and shape, so repeat with **several window sizes and aspect ratios**:
  - tens of thousands of CNN evaluations per image.
  - Boxes are only as precise as the grid, and most windows show background.
- **Conclusion:** too slow and wasteful; we need **fewer, smarter candidates**.

Region Proposals and Selective Search

- A **region proposal** is a candidate box that is **likely** to contain some object, whatever its class.
- **Selective search** (classical, not learned):
    - start with many tiny regions, **repeatedly merge** the most similar neighbours (similar colour, texture, size); every
        - **merged region is a proposal.**
- Produces ≈ **2,000 proposals** per image, covering objects at all scales.
- Compared with 10^4-10^5 sliding windows this is a **big reduction**, and proposals follow object boundaries better.
- But it runs on the CPU and takes seconds per image.

R-CNN (2014): Region-based CNN
Regions with CNN Features



- Each **proposal** is **cropped and resized (warped)** to the CNN input size, then passed **separately** through the CNN.
- One **Support Vector Machine (SVM)** per class scores the features;
    - a **bounding-box regressor** fixes the box.
- **Big accuracy gain** over hand-made features on Pascal VOC (Visual Object Classes) dataset.
- **Problem**: 2000 separate CNN passes per image: ≈ **47 s per image** with VGG-16 (Fast R-CNN paper), and three separate training stages.

Fast R-CNN and Faster R-CNN

Fast R-CNN

Run the CNN **once** on the whole image; **RoI pooling** extracts a fixed-size feature for each proposal. Much faster; proposals are still slow.

Faster R-CNN

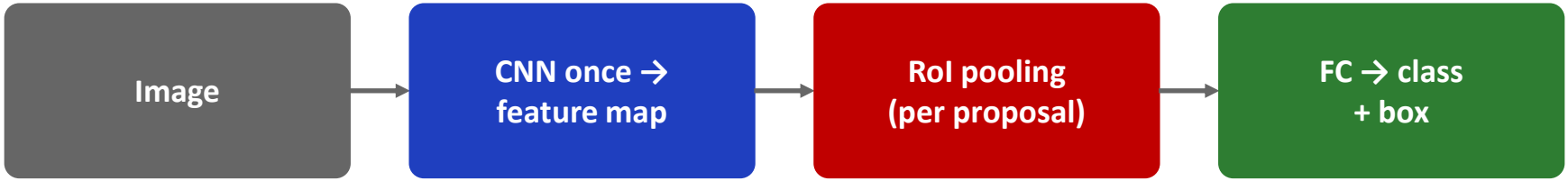
A **Region Proposal Network (RPN),** which is a small conv network sliding over the feature map**,** predicts proposals from the shared feature map using **anchors**.

An **anchor** is a reference box (fixed w_a, h_a) centred at every feature-map location

≈ **5 fps** with VGG-16 (vs 0.02 fps for R-CNN)

Still two stages process: (1) propose regions, (2) classify and refine them.

Fast R-CNN (2015): Share the Computation



- Run the CNN **once on the whole image**, then **cut each proposal's region out of the shared feature map**.
- **RoI (Region of Interest) pooling**: divide the RoI into a fixed grid/cell (e.g. 7×7) and **max-pool each cell**.
  - Any RoI size → a fixed 7×7 feature, ready for FC layers.
  - Example: a **14×21** RoI on the feature map → **7×7** grid, each cell **2×3** → max over **6 values**.
- Class and box are trained **together** (multi-task loss $L = L_{cls} + \lambda \cdot Lbox$).
- Roughly **100× faster at test time** than R-CNN, but proposals (selective search) are still the bottleneck.

Two-Stage vs. One-Stage Detectors

	Two-stage (R-CNN family)	One-stage (YOLO, SSD, RetinaNet)
Pipeline	1) propose regions, 2) classify and refine each	Predict boxes and classes directly from the feature map in one pass
Accuracy	Historically higher	Now comparable, especially for modern versions
Speed	Slower (≈ 5-15 fps originally)	Real-time (30-100+ fps)
Strength	Careful per-region reasoning	Simple, fast, easy to deploy on video / edge devices
Weakness	Complex, slower	Class imbalance harder (fixed by focal loss, better assignment)

Next lecture: YOLO treats detection as a single regression problem.

Non-Maximum Suppression (NMS)

- **Problem**: many nearby anchors / cells fire on the **same object**, giving several overlapping boxes.
- **NMS algorithm** (done **per class**):
    1. Discard boxes with score below a **confidence threshold** (e.g. 0.25).
    2. Sort the remaining boxes by score, descending.
    3. Take the top box and **keep** it.
    4. Delete every other box with **IoU > t_{NMS}** (e.g. 0.5) with the kept box.
    5. Repeat from step 3 with the boxes left, until none remain.
- t_{NMS} trades off:
    - **too low** removes neighbouring real objects;
    - **too high** leaves duplicates.

Evaluating Detectors: AP and mAP

- **For one class, rank detections by score**;
    - sweep the threshold to get a **precision-recall curve**

- **AP** = area under the PR curve
- **mAP** = mean AP over classes
- **mAP@0.5**: TP needs IoU ≥ 0.5
- **COCO mAP@[0.5:0.95]**: average over IoU thresholds 0.5, 0.55, …, 0.95 (stricter, rewards tight boxes)

Average Precision (AP): Worked Example

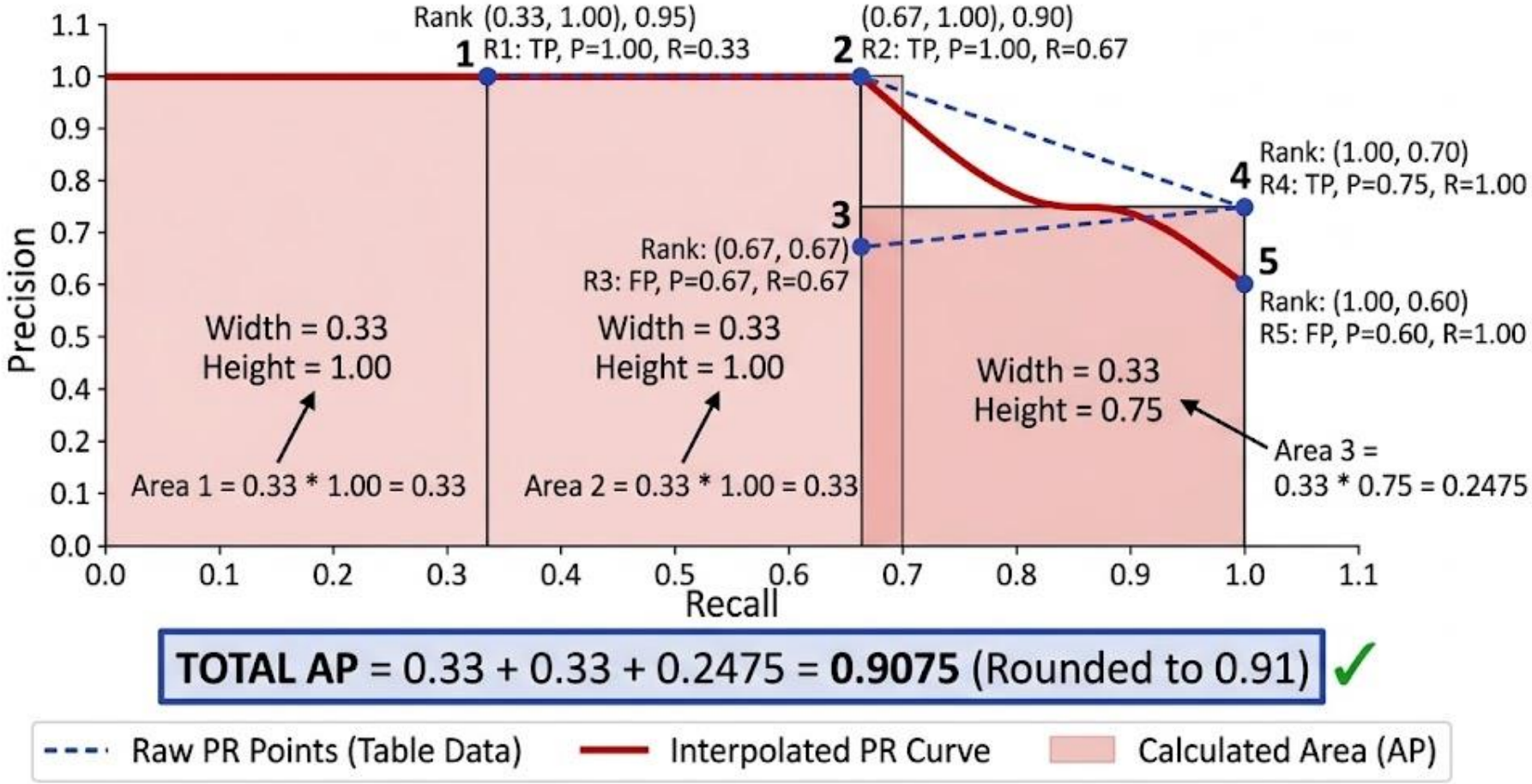
- One class, **3 ground-truth objects**.
  - **Detections sorted by score**;
  - **precision** = $\frac{TP}{(TP+FP)} = \frac{TP}{Total\ Detections\ Evaluated}$ so far,
  - **recall** = $\frac{TP}{(TP+FN)}$ = Percentage of total ground-truth objects found so far = $\frac{TP}{3}$.

Note: AP = Area under the PR curve

Rank/Detection	Score	Result	Precision	Recall
1	0.95	TP	1/1 = 1.00	1/3 = 0.33
2	0.90	TP	2/2 = 1.00	2/3 = 0.67
3	0.80	FP	2/3 = 0.67	0.67
4	0.70	TP	3/4 = 0.75	3/3 = 1.00
5	0.60	FP	3/5 = 0.60	1.00

- **AP** = area under the precision-recall curve, using at each recall the **best precision at that recall or higher**.
  - Recall 0 → 0.33: precision 1.00;
  - Recall 0.33 → 0.67: precision 1.00;
  - Recall 0.67 → 1.00: precision 0.75.
- AP = 0.33×1.00 + 0.33×1.00 + 0.33×0.75 ≈ **0.91**.

Meaning for each recall bin, you take the **maximum precision achieved at that recall level or any higher recall level**



TOTAL AP = 0.33 + 0.33 + 0.2475 = **0.9075** (Rounded to 0.91) ✓

Raw PR Points (Table Data) — Interpolated PR Curve — Calculated Area (AP)

Datasets

Dataset	Classes	Size / note
PASCAL VOC	20	2007-2012; ≈ 10-20k images; classic benchmark (mAP@0.5)
MS COCO	80	≈ 330k images, ≈ 1.5M object instances; standard benchmark; also masks
Open Images	600 boxable	≈ 1.9M images, ≈ 15M boxes (version 4)
Your own data	your classes	Label with tools such as CVAT, Label Studio or Roboflow

Why One-Stage Detectors? Toward YOLO

- **Real-time need**: video at 30 fps leaves **≈ 33 ms per frame** for everything; robots and cars need the same.
- Two-stage pipelines have **two networks, many hyper-parameters and variable cost** per image.
- Idea (YOLO, 2016): **drop proposals**. Divide the image into a grid and let each cell predict boxes and classes **directly in one forward pass**.
- Same tools reappear: e.g., **anchors**, **IoU**, **NMS**, **mAP**.

Check Yourself: Questions and Answers

Question	Answer
IoU of (0,0,4,4) and (1,1,5,5)?	intersection 9, union 23 → 0.39
Why did Faster R-CNN replace selective search?	Learned proposals share the CNN features: faster and trainable end to end.
What does RoI pooling output for any RoI size?	A fixed-size feature (e.g. 7×7) via grid max-pooling.
What does COCO mAP average over?	Classes and IoU thresholds 0.50-0.95.

Lecture 8 Summary

- Detection = **class + box + score** for a variable number of objects.
- **IoU** measures overlap and defines TP / FP / FN; **NMS** removes duplicates.
- **R-CNN → Fast → Faster R-CNN**: shared features, RoI pooling, learned proposals (RPN).
- **AP / mAP** summarize precision-recall across classes and IoU thresholds.
- Next: **YOLO**, detection in a single pass.